Pizza Sales Data Analysis

SQL-Based Business Insights & KPI Analysis

Prepared by: Papai Chakraborty
Role: Data Analyst Project
Tools Used: SQL Server (SSMS)
Date: April 2026

SQL Queries - Pizza Sales

A. KPI's

1. Total Revenue:

```
SELECT SUM(total_price) FROM pizza_sales;
```

Results | Messages

	Total_Revenue
1	817860.05083847

2. Average Order Value (AOV):

```
SELECT
    SUM(total_price) AS total_revenue,
    COUNT(DISTINCT order_id) AS total_orders,
    ROUND(SUM(total_price) / COUNT(DISTINCT
order_id), 2) AS AOV
FROM pizza_sales;
```

Results | Messages

	total_revenue	total_orders	AOV
1	817860.05083847	21350	38.31

- **Note** - Since each row represents a pizza and not an order, I used total revenue divided by distinct order count to correctly calculate AOV.

3. Total Pizzas Sold:

```
SELECT SUM(quantity) AS Total_Pizzas_Sold FROM
pizza_sales;
```

Results | Messages

	Total_Pizzas_Sold
1	49574

4. **Total Orders:**

```
SELECT COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales;
```

Results | Messages

	total_orders
1	21350

5. **Average Pizzas Per Order:**

```
SELECT CAST(CAST(SUM(quantity) AS DECIMAL(10,2))
/
CAST(COUNT(DISTINCT order_id) AS DECIMAL(10,2))
AS DECIMAL(10,2)) AS Average_Pizzas_Per_Order
FROM pizza_sales;
```

Results | Messages

	Average_Pizzas_Per_Order
1	2.32

B. Daily Trend for Total Orders:

```
SELECT DATENAME(DW,order_date) AS order_day ,
COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY DATENAME(DW,order_date)
ORDER BY total_orders DESC;
```

Results | Messages

	order_day	total_orders
1	Friday	3538
2	Thursday	3239
3	Saturday	3158
4	Wednesday	3024
5	Tuesday	2973
6	Monday	2794
7	Sunday	2624

C. Hourly Trend for Orders:

```
SELECT DATEPART(HOUR,order_time) AS
Hourly_orders, COUNT(DISTINCT order_id) AS
Total_orders
FROM pizza_sales
GROUP BY DATEPART(HOUR,order_time)
ORDER BY Hourly_orders;
```

Results | Messages

	Hourly_orders	Total_orders
1	9	1
2	10	8
3	11	1231
4	12	2520
5	13	2455
6	14	1472
7	15	1468
8	16	1920
9	17	2336
10	18	2399
11	19	2009
12	20	1642
13	21	1198
14	22	663
15	23	28

D. Percentage of Sales by Pizza Category:

```
SELECT pizza_category,SUM(total_price) AS
Category_Revenue,
ROUND(SUM(total_price) * 100/
SUM(SUM(total_price)) OVER(),2) AS
Sales_Percentage
FROM pizza_sales
GROUP BY pizza_category;
```

Results | Messages

	pizza_category	Category_Revenue	Sales_Percentage
1	Classic	220053.100021362	26.91
2	Chicken	195919.5	23.96
3	Veggie	193690.451004028	23.68
4	Supreme	208196.99981308	25.46

E. Percentage of Sales by Pizza Size:

```
SELECT pizza_size,SUM(total_price) AS
Size_Revenue,
ROUND(SUM(total_price) * 100/
SUM(SUM(total_price)) OVER(),2) AS
Sales_Percentage
FROM pizza_sales
GROUP BY pizza_size;
```

Results | Messages

	pizza_size	Size_Revenue	Sales_Percentage
1	L	375318.701004028	45.89
2	XXL	1006.6000213623	0.12
3	M	249382.25	30.49
4	XL	14076	1.72
5	S	178076.49981308	21.77

F. Total Pizzas Sold by Category:

```
SELECT pizza_category , SUM(quantity) AS
Total_Quantity_Sold
FROM pizza_sales
GROUP BY pizza_category
ORDER BY Total_Quantity_Sold DESC;
```

Results | Messages

	pizza_category	Total_Quantity_Sold
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

G. Top 5 Best-Selling Pizzas:

```
SELECT TOP  5 pizza_name ,SUM(quantity) AS
Total_Quantity
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity DESC;
```

Results | Messages

	pizza_name	Total_Quantity
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

H. Bottom 5 Worst-Selling Pizzas:

```
SELECT TOP  5 pizza_name ,SUM(quantity) AS
Total_Quantity
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity ASC;
```

Results | Messages

	pizza_name	Total_Quantity
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Pizza	950
5	The Soppressata Pizza	961

I. Revenue Contribution by Pizza Type:

```
SELECT pizza_category ,
ROUND(SUM(total_price),2) AS Revenue
FROM pizza_sales
GROUP BY pizza_category;
```

Results | Messages

	pizza_category	Revenue
1	Classic	220053.1
2	Chicken	195919.5
3	Veggie	193690.45
4	Supreme	208197

J. Top 5 Highest Value Orders:

```
SELECT TOP 5 order_id,ROUND(SUM(total_price),2)
AS order_value
FROM pizza_sales
GROUP BY order_id
ORDER BY order_value DESC;
```

Results | Messages

	order_id	order_value
1	18845	444.2
2	10760	417.15
3	1096	285.15
4	6169	284
5	740	280.95

Key Insights & Business Recommendations

- The Average Order Value (AOV) is 38.31, indicating moderate customer spending per order. This suggests an opportunity to increase revenue through upselling strategies such as combo deals and add-ons.
- Order volume peaks on Friday, Saturday, and Thursday, indicating higher customer demand toward the end of the week and during weekend
- **12 pm , 1 pm , 5 pm , 6 pm , 7 pm Order Volume peaks.**

Actions we can take

- Increase **staffing & inventory** on these days and Times.
- Launch **weekend offers or combos**
- Run **targeted promotions (Thu-Sat)**
- Extend **delivery capacity during peak days** and peak Hours.

- Sales analysis shows that Classic and Supreme categories generate the highest revenue contribution, with Large and Medium sizes being the most preferred among customers.
- Based on quantity sold, the Classic Deluxe Pizza ranks as the best-performing product, whereas the Brie Carre Pizza shows the lowest demand among all items.

Actions we can take

For Best Seller

- **Feature it as "Best Seller" on menu**
- **Put it on homepage / top position (online apps)**
- **Create combo offers (Classic Deluxe + drink)**
- **Slightly increase price (carefully) to boost profit**

For Worst Seller

- **Change recipe / taste**
- **Adjust pricing**
- **Rebrand name (important sometimes)**
- **Offer discounts or combos**
- **"Try something new" campaigns**
- **Or Remove it from menu.**

Papai Chakraborty